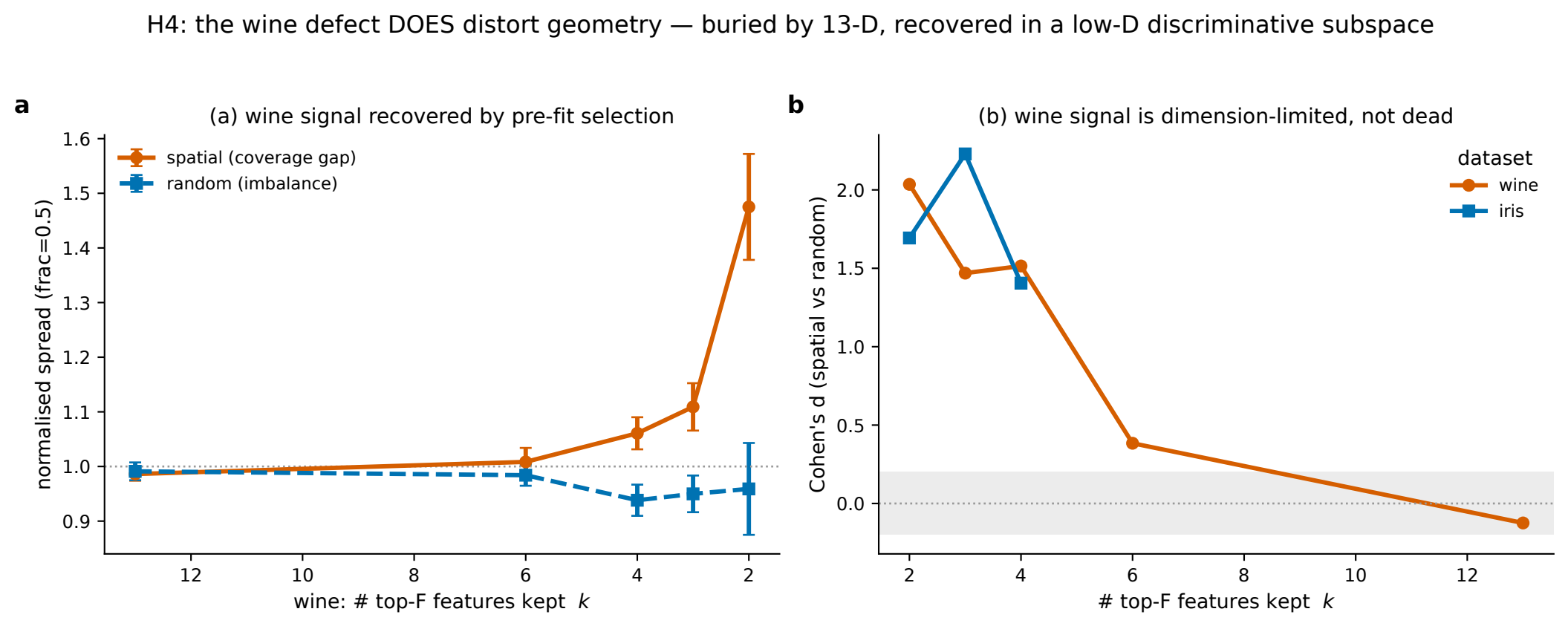
H4: the wine defect DOES distort geometry — buried by 13-D, recovered in a low-D discriminative subspace
a
(a) wine signal recovered by pre-fit selection
spatial (coverage gap)
random (imbalance)
normalised spread (frac=0.5)
1.6
1.5
1.4
1.3
1.2
1.1
1.0
0.9
12
10
8
6
4
2
wine: # top-F features kept k
b
(b) wine signal is dimension-limited, not dead
dataset
wine
iris
Cohen's d (spatial vs random)
2.0
1.5
1.0
0.5
0.0
2
4
6
8
10
12
top-F features kept k